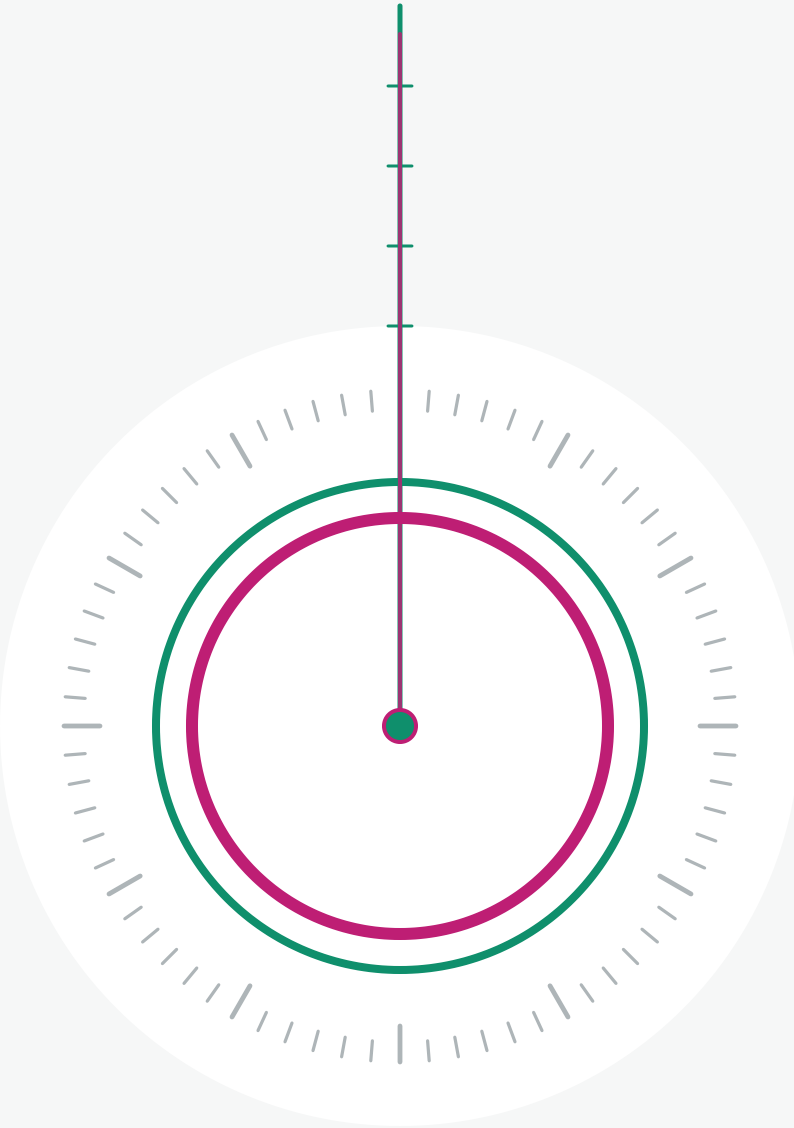
DATUM LINE

# Every lab's cells. One "normal".

Membership to a shared repository
of every lab's control cells.



**Kat Titterton**
Cofounder, CSO, San Francisco



resume ↗

**8**
**peer-reviewed papers**
two in Nature Communications ↗

**2**
**patent applications**
filed at Axiom Bio

**100,000+**
**compounds imaged**
two Cell Painting datasets ↗

# Kat Titterton

## WORK

**2023–2026** Axiom
cell imaging, AI

**2022–2023** insitro
automated screening

**2021–2022** MIT
expansion microscopy

**2018–2021** abbvie
neuron assays

**2017–2018** MGH
electrophysiology

## EDUCATION

**2013–2017** COLGATE BA
neuroscience, math

## MY WORK

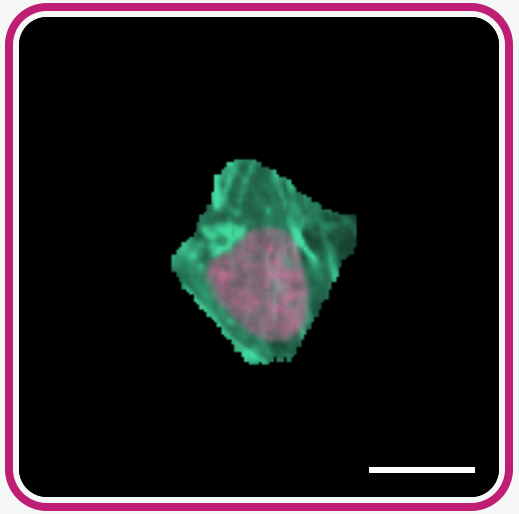
1. Cell Painting in primary human hepatocytes
   *Cell Systems* 2026, Ewald, Broad Institute
2. Counting cells predicts bioactivity benchmarks
   *Nat Commun* 2026, Seal, Univ. of Cambridge
3. Cell Painting of hepatotoxic botanicals
   *bioRxiv* 2026, Ollitrault, EMBL-EBI
4. An antibody to pathological tau
   *Mol Neurodegen Adv* 2026, Yanamandra, AbbVie
5. Capturing synaptic proteopathic seeds
   *iScience* 2026, Sebastian, AbbVie
6. A neuronal tau aggregation assay
   *J Biol Chem* 2025, Wu, AbbVie
7. Multiplexed imaging of brain nanostructures
   *Nat Commun* 2024, Kang, MIT
8. Tau aggregation in human iPSC neurons
   *iScience* 2022, Manos, AbbVie
9. Neurosteroid photolabeling of $GABA_A$ receptors
   *Eur J Med Chem* 2019, Wu, U Illinois Chicago

# Do my cells look “normal”?

“normal” = published images + past data

asked @ five benches
… answered @ none

“Normal” compared with whom?
Each lab’s own checks can pass while labs still disagree (slides 03 and 04, archive A13).



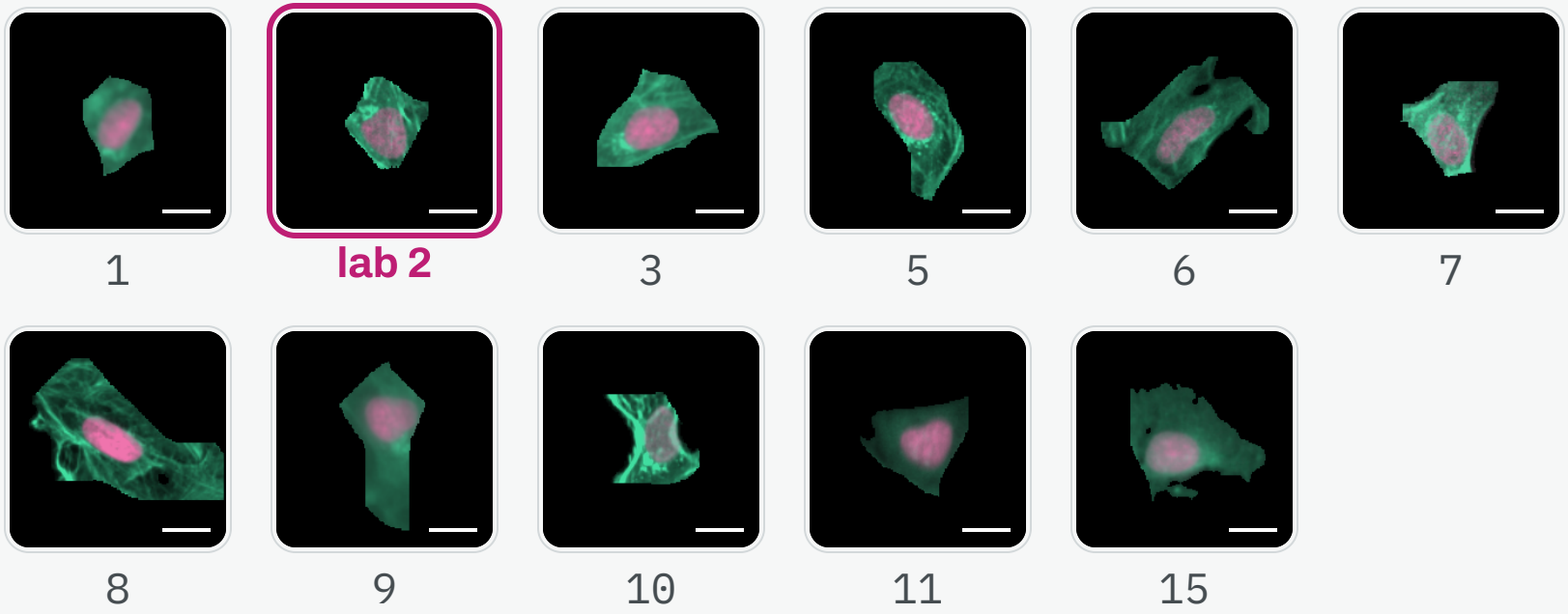
my cells, lab 2

DMSO, 20 µm bar

# Each lab judges “normal” alone.

$28B a year of US lab research others cannot reproduce [6]

36% ≈ $10B, cells and reagents | 64%, other causes




WHAT IT COSTS

**$100,000s**

a screen on changed cells [28]:

about $50k a run, 3 runs a week

**months**

asking around, or re-running

[28]

**Why now** 2023 1.8B public cells 2025 $30M per cell map now AI reads cells

why now: 1.8B public cells, JUMP [7, 25]; FDA: fewer animal tests [9]; $30M per cell map, Roche to Recursion, Q3 2025 results [11]; AI reads cells (A3)

# One lab, or eleven?

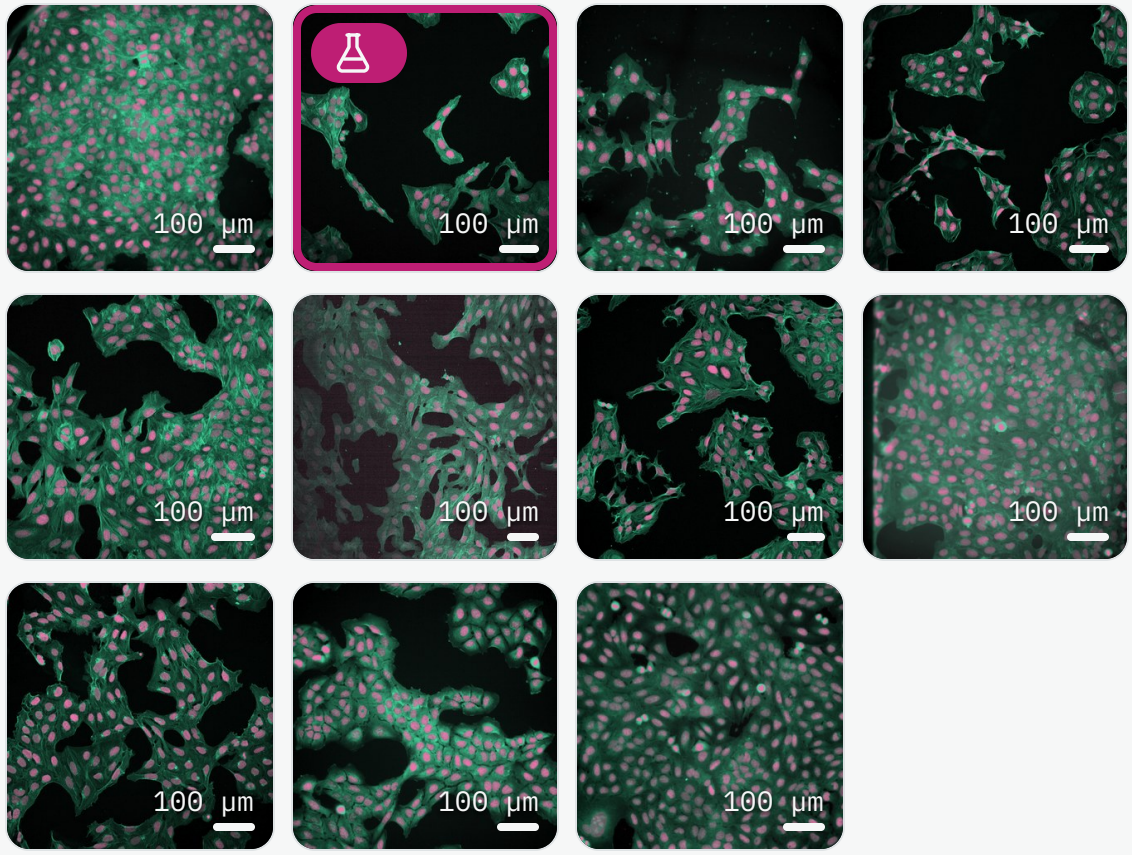



**fig 1** Cell Painting, images: DNA channel, classifier: 3,773 numbers per well from all channels, 93,228 wells, 11 labs (a lab = one JUMP data source, its data-generating centre), plates held out, bars 100 µm, JUMP [7], archive A2, A1, archive A5

## same cells

U2OS: human bone cancer cells

osteosarcoma (bone cancer) from a 15-year-old's tibia, described in 1967 [26]

the one line in JUMP's full dataset, so every lab here grew it [27]

chosen over A549 by a slim margin, for its public data [27]

named in 11 of the Cell Painting Gallery's 28 datasets [17]

a cancer line: "normal" means its DMSO baseline, not healthy tissue

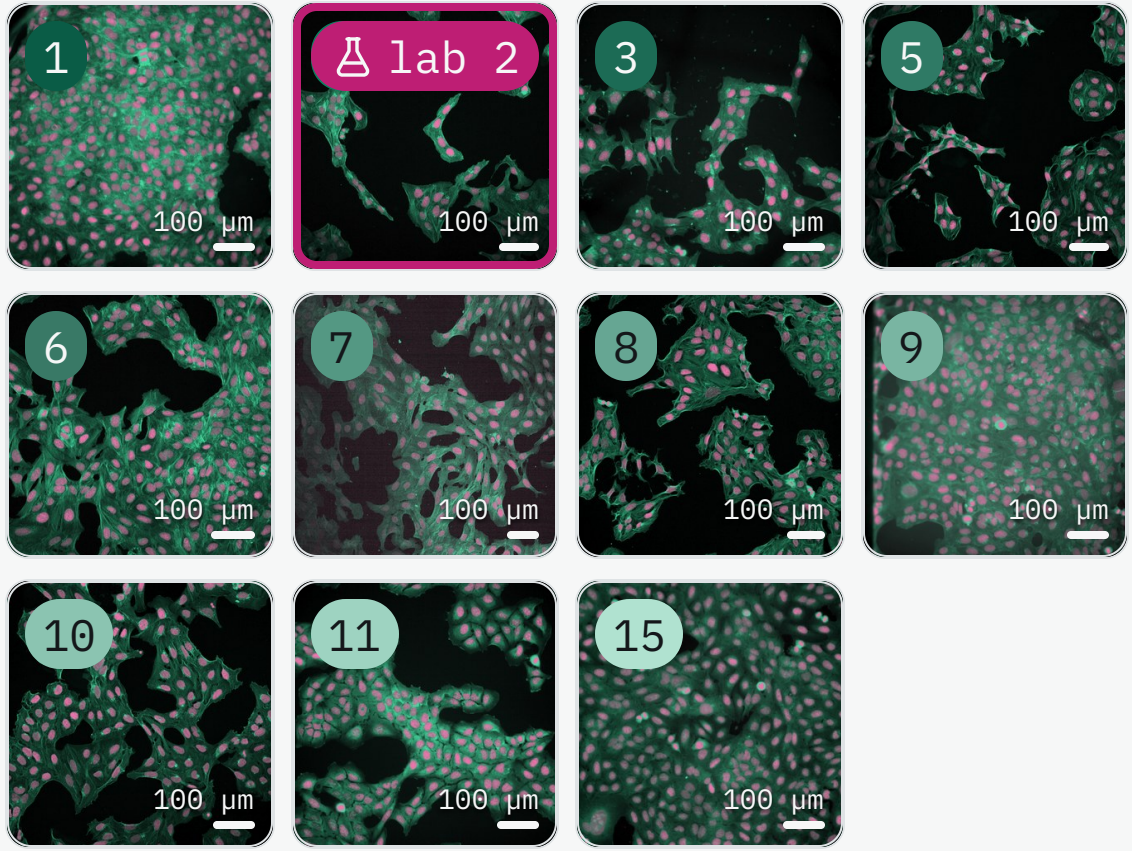
## same protocol

Cell Painting, one protocol at every site [7, 27]

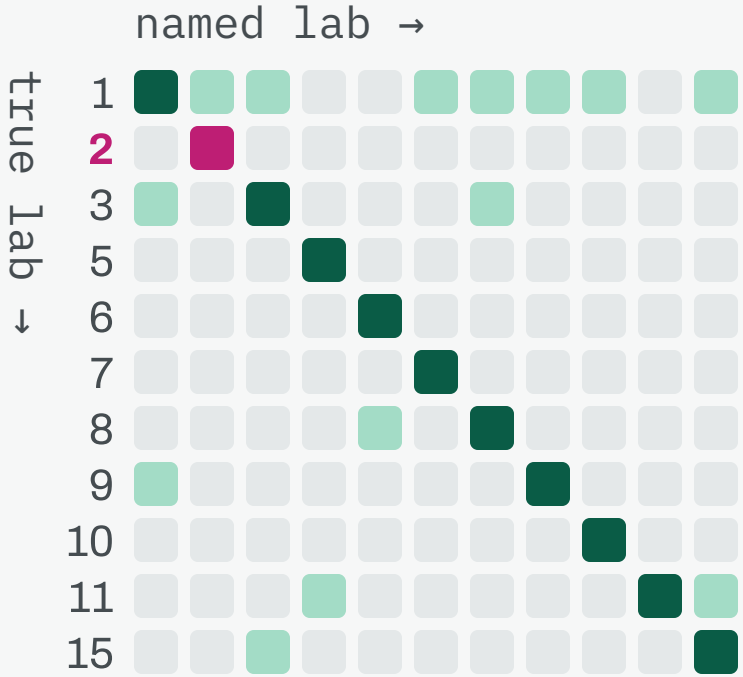
## no drug

DMSO only, the drugs' solvent [7]

# Could anyone tell?
# A classifier can.



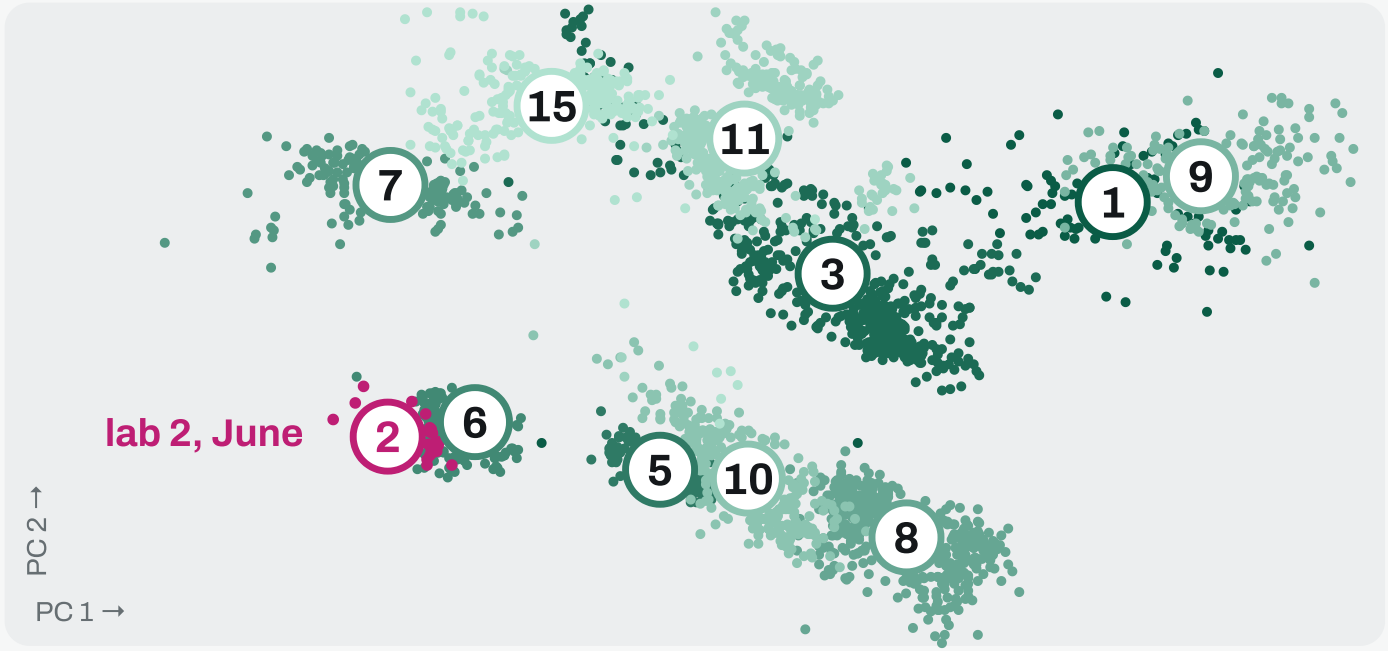

**99.9%** right lab, of 11
chance: 1 in 11




shade: the share of a row's wells, light grey none to dark all; 93,228 wells, results/e1

**fig 1** Cell Painting, images: DNA channel, classifier: 3,773 numbers per well from all channels, 93,228 wells, 11 labs (a lab = one JUMP data source, its data-generating centre), plates held out, bars 100 µm, JUMP [7], archive A2, A1, archive A5

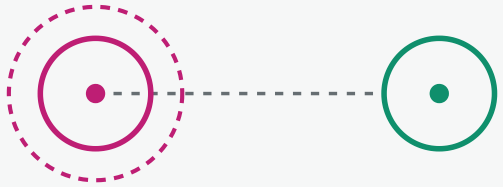
# Eleven labs.
# Eleven “normals”.




one well  7 a lab’s “normal”  closer = more alike

**6×**

between two labs



**lab 2**  another lab

- one well: ~220 cells
- 1× wells on one plate
- 1.6× a lab’s batches

dots: DMSO wells, 8 per plate, one in three drawn; PCA of 15,352 wells, for the eye (distances use all 3,773 numbers); plates: 384 wells, 1,536 in labs 1 and 9; 6×: two labs’ centres sit 6.2 plate spreads apart on average, a plate spread being a well’s mean distance from its own plate’s centre, all 3,773 numbers (results/e13); JUMP [7]; archive A2, A1, archive A13

# Send the controls.
# Know it’s “normal” before the screen.




## Member check

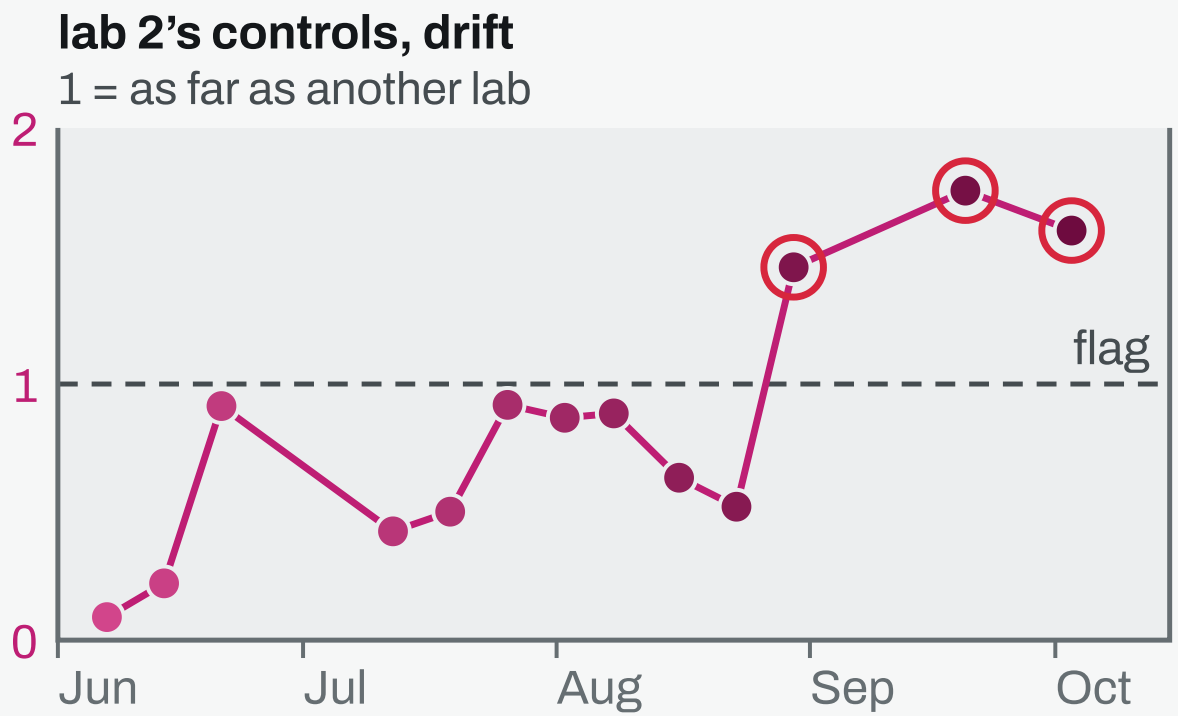
a new batch, in all features

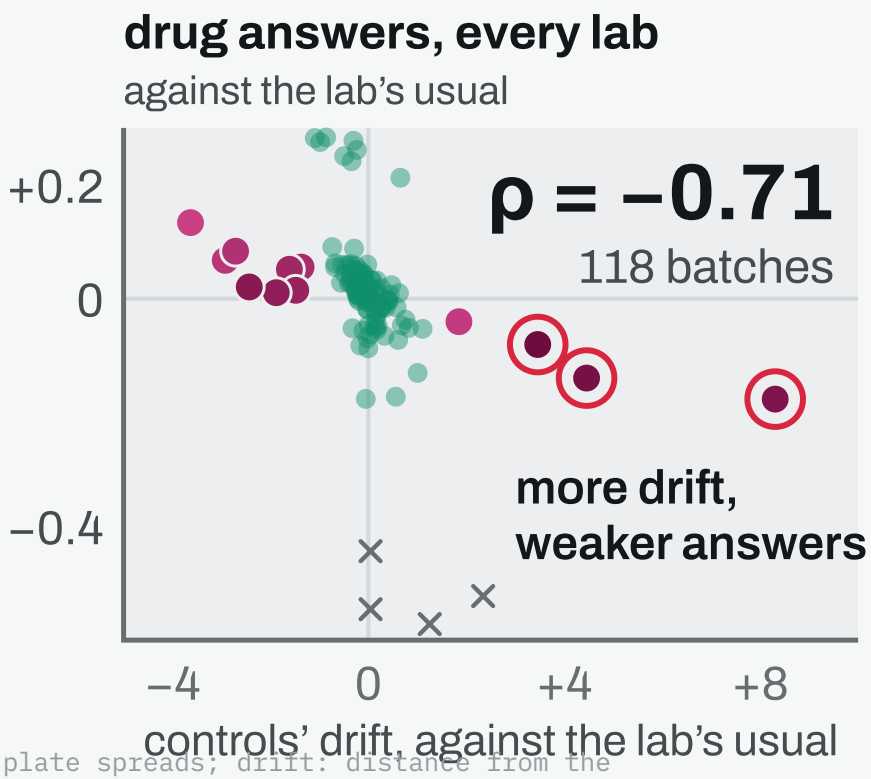
| | | |
|---|---|---|
| own lab | ■ | 2.8× |
| next lab | ■ | 4.4× |

| | |
|---|---|
| nearer its own lab | ✓ |
| drugs match | ✓ |
| grew as usual | ✓ |

the member check, replayed on public data; closest lab in all 3,773 numbers: lab 6, 4.4 within-plate spreads away; known drugs worked; agreement with other labs 0.70 measured, 0.77 on the map; 188 cells per well, in its usual range; JUMP [7], results/e9, e11, e12; archive A11, A6

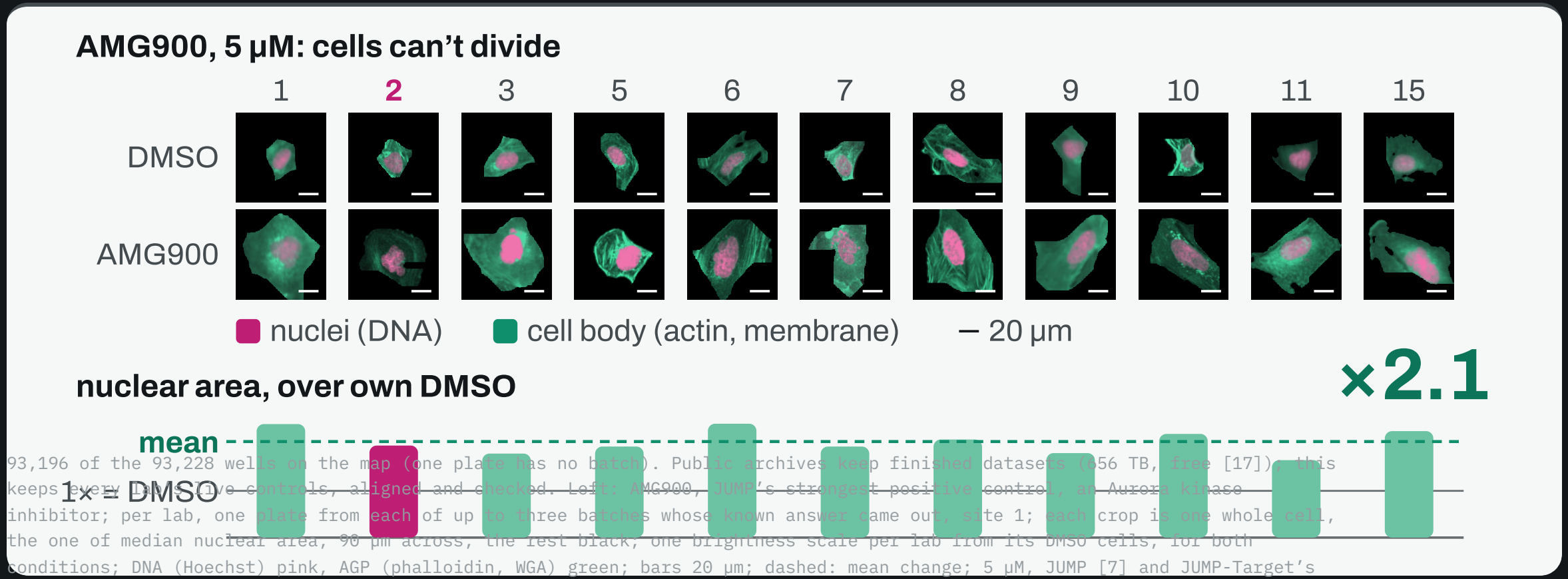
# Controls catch drift before any drug is read.



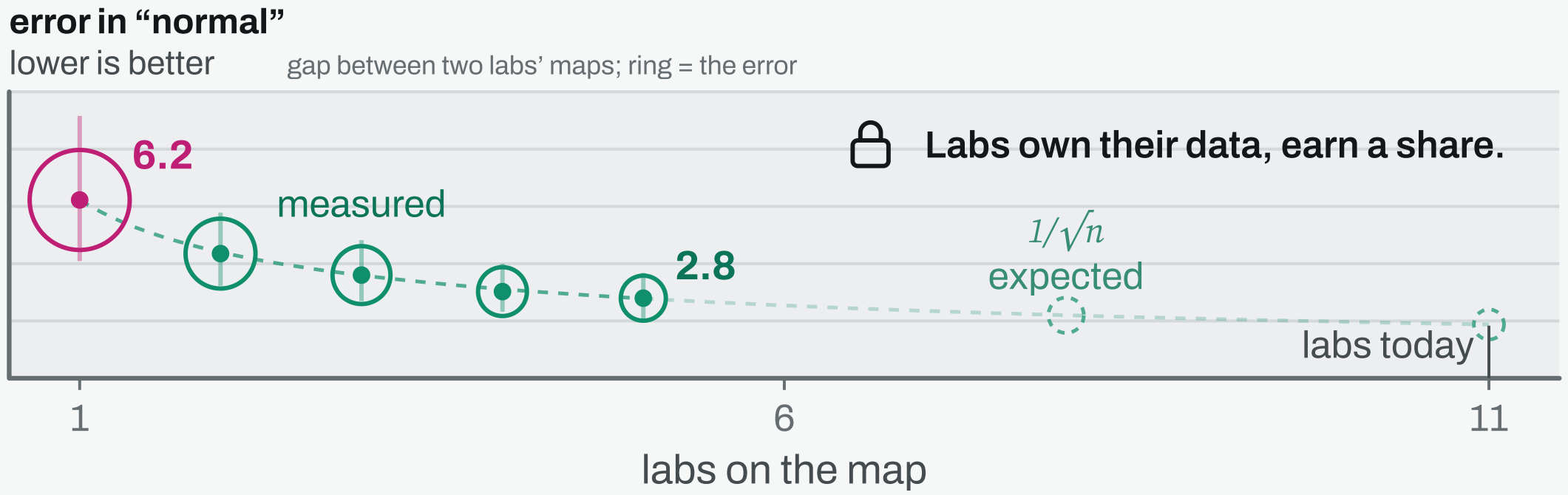

lab 2, 13 batches; drift: control wells' distance from the first batch, in within-plate spreads; drift: distance from the first batch over distance to the nearest other lab's centre, so 1 is the flag line (most plates nearer another lab than the first batch, e9); right: every batch's drift and match minus its lab's mean, Spearman over all labs, permutation p = 0.0005, crosses: positive controls failed (results/e10); archive A6; JUMP [7]

# The product is the data.
# One drug, one shared change.




93,196 of the 93,228 wells on the map (one plate has no batch). Public archives keep finished datasets (656 TB, free [17]); this keeps every lab's live controls, aligned and checked. Left: AMG900, JUMP's strongest positive control, an Aurora kinase inhibitor; per lab, one plate from each of up to three batches whose known answer came out, site 1; each crop is one whole cell, the one of median nuclear area, 90 µm across, the rest black; one brightness scale per lab from its DMSO cells, for both conditions; DNA (Hoechst) pink, AGP (phalloidin, WGA) green; bars 20 µm; dashed: mean change; 5 µM, JUMP [7] and JUMP-Target's recommended dose (results/e18). The seven known drugs' agreement across labs, alone against aligned: A4 (results/e12). JUMP [7], A5, A6, A11

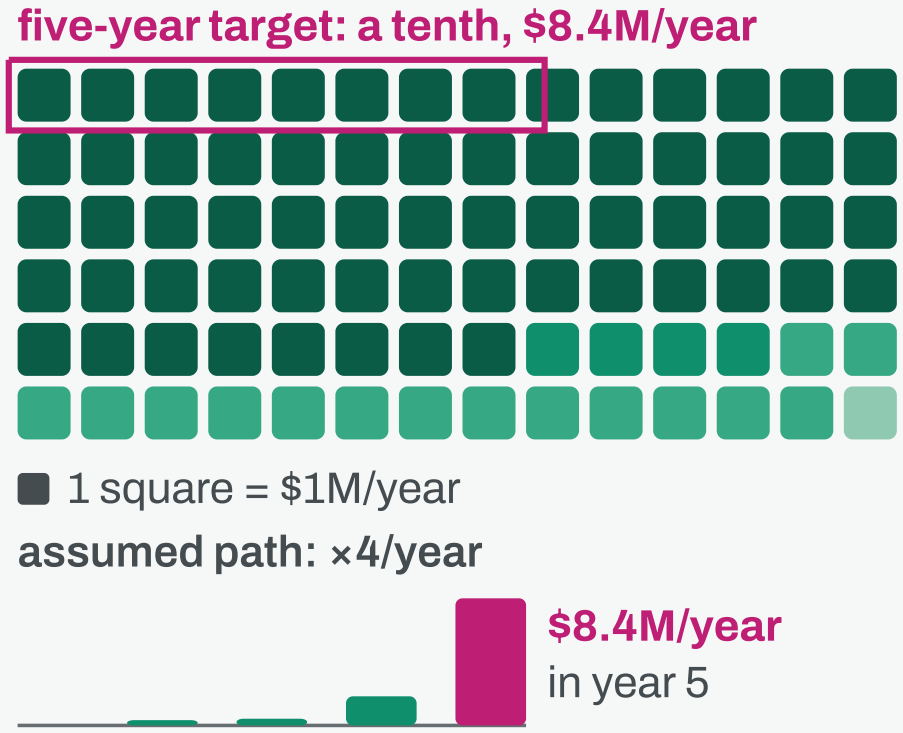
# Every lab that joins sharpens “normal” for everyone.




**Go-to-market** 3 partner labs join (A9) → screening labs subscribe (A10) → AI teams subscribe by API (A5)

**fig 5** error: distance between two maps from different labs, in within-plate spreads; dot: mean of 400 draws, bar: 10th to 90th percentile; measured to five labs, dashed beyond; 1,871 plates; JUMP [7]; results/e13

# Every likely buyer: $84M/year.
# Five-year target: a tenth.




| | | |
|---|---|---|
| screening labs<br>**$10k/line/quarter** | **$64M**<br>400 labs × 4 lines |
| big screening panels<br>**$150k/year** | **$3.75M**<br>25 panels |
| buyers of cell data<br>**$25k/dataset** | **$15M**<br>150 buyers × 4 datasets |
| benchmark partners<br>**$100k/year** | **$1M**<br>10 partners |

academic labs: free  AI teams: by API

labs are the map; API pricing not yet set

**beyond the base case**

AI cell data: a $1.8B push [12]

irreproducible research: $28B/year [6]

**fig 6** base case of A10: every count and price an assumption until a count replaces it; low case $9.7M/year; academic labs join free and are the map, not the revenue; AI usage by API not priced; why now is on 02, go-to-market on 06

# Each tool sees one lab.
# Datum Line sees every lab.

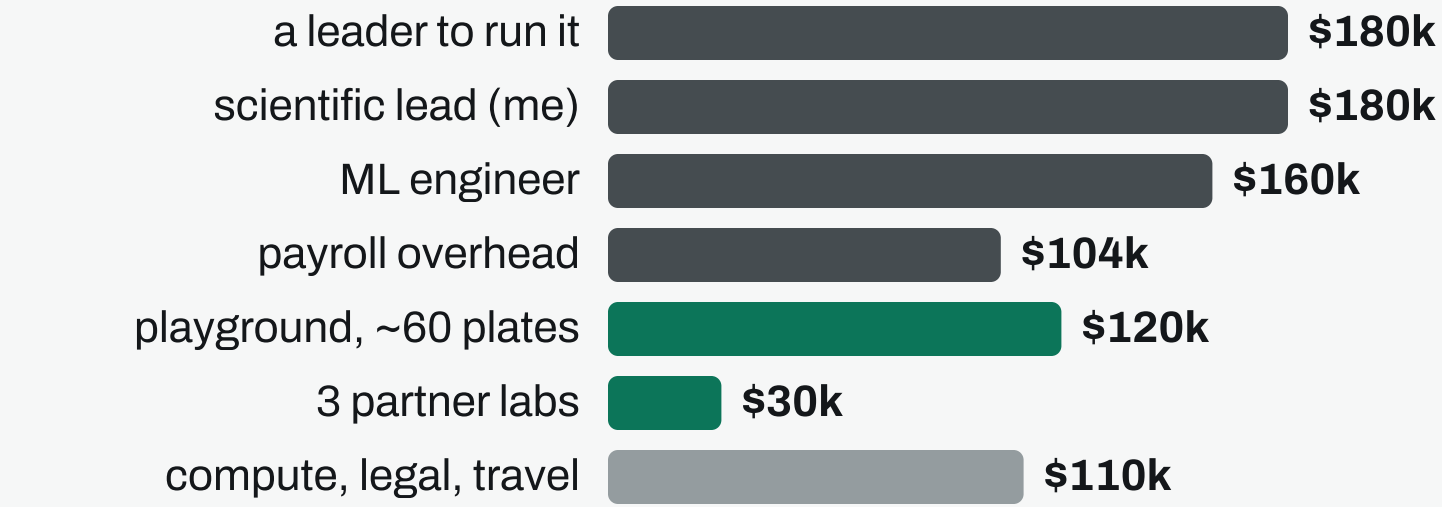
| | measure cells | normalise plates | reference data | every lab, over time |
|---|---|---|---|---|
| **CellProfiler**<br>image analysis, open source [32] | ✓ | | | |
| **Harmony**<br>Revvity's Opera Phenix software [33] | ✓ | | | |
| **pycytominer**<br>profiles, each plate to its own DMSO [34] | | ✓ | | |
| **Genedata Screener**<br>screen QC and normalisation [35] | | ✓ | | |
| **JUMP, Recursion**<br>reference data at scale [7, 11] | | | ✓ | |
| **Datum Line**<br>every lab's controls, one repository | | | | ✓ |

tools as a phenotypic screen runs them: CellProfiler [32]; Harmony, Revvity [33]; pycytominer [34]; Genedata Screener [35]; JUMP [7], Recursion [11]. A check: the tool does the step, from its own documentation. The wider landscape (cell-line checks, vendor certificates): docs/landscape.html

Datum Line hosts every lab's raw control images, and runs on them.

Why not them? JUMP is a public archive [7], Recursion maps its own cells [11], Revvity sells the microscope [33]. Landscape [3].

# Raising $1.8M pre-seed: first paying members, then the seed.




$884k plan + $884k contingency = $1.77M (A9)

proposed; milestones in order, not dated; every line an assumption (A9)

TODAY

- **11 labs, 129 batches mapped** from public data (JUMP [7], results/e9); no revenue yet

18 MONTHS

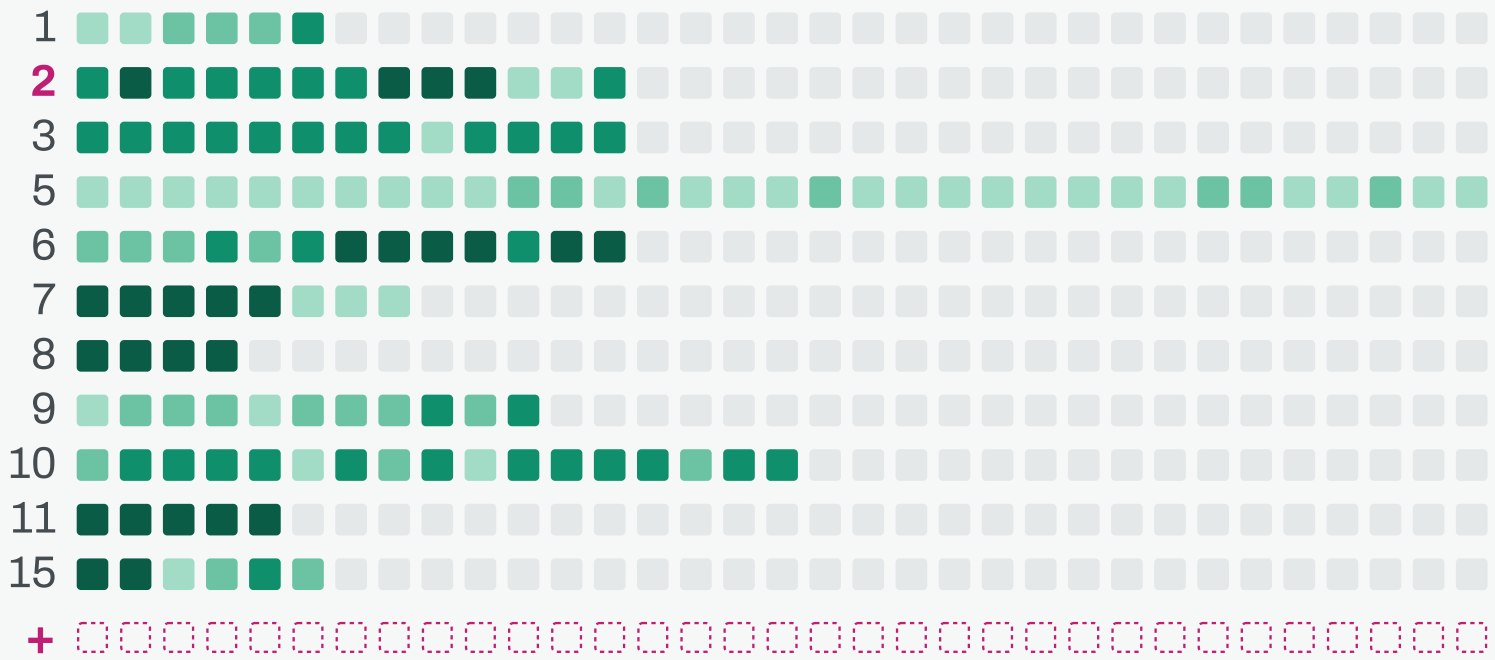
1. **three partner labs, the first outside JUMP**
2. **first paying member**
3. **benchmark playground**
4. **model for any image**

→ seed round

roadmap ↗

# Every plate imaged, placed on the map.

Labs share control images, free.
Each sees its “normal” next to everyone else’s.




**129** batches today; the open row, yours

Less More plates

**fig 7** one square: one batch, in run order, shaded by plates at the quartiles; a median of 10 plates (1 to 58); 1,871 plates, JUMP [7], results/e9; A5, A8, A10, A11

## $1.8M

pre-seed:
first members, then the seed, 18 months, A9

LOOKING FOR

- **1** leader, to run it
- **2** benchmark partners
- **3** partner labs, bench time

**Have control images?**
**Talk to me.**

open code ↗ all documents ↗

# The rest, for questions.

Every number: the code here or a cited source.
All 25 boards: archive.html
Documents: white paper, product vision, roadmap, landscape

**A1 Why not batch-correct it away?**
erases most of the lab fingerprint

**A2 Does it change the result?**
same drug, different answer

**A3 Does it work without stains?**
bet on brightfield

**A4 What does the map buy?**
same drug, closer between labs

**A5 Who pays?**
labs join free, subscribers pay

**A6 What is private?**
control wells leave, nothing else does

**A7 What is the machine learning?**
image in, position and correction out

**A8 What about biosecurity?**
cells show their lab, labels can lie

**A9 How much, and what for?**
planned $0.9M × 2 = $1.8M asked

**A10 How big?**
count × price, three streams

**A11 Why would a lab share?**
owned by the lab, verified by anyone

# The usual correction erases most of the lab's fingerprint.

JUMP records the left column; nobody records the middle.

RECORDED IN JUMP

- lab
- batch date
- plate
- plate type
- well position
- microscope
- confocal or widefield
- laser or LED
- objective NA
- fields per well
- filter set
- pixel size
- brightfield planes
- CellProfiler version

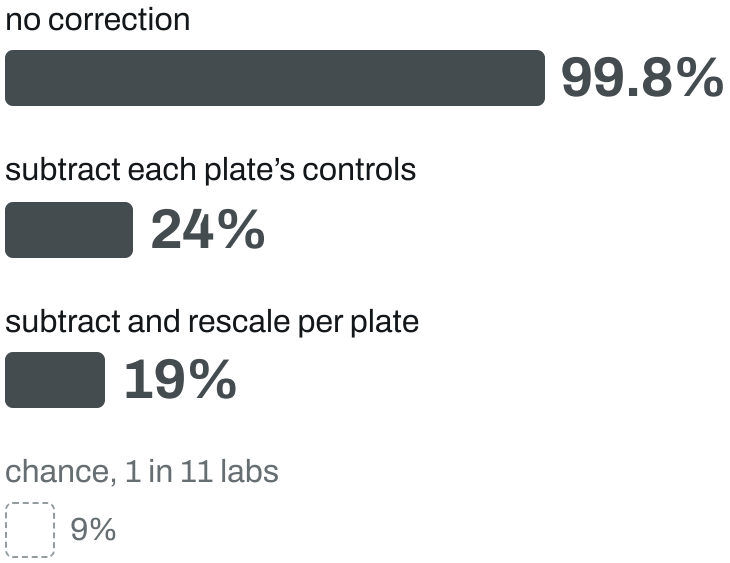
in every image file too:
image size, stored bit depth

NOT IN ITS PUBLIC METADATA

- DMSO % per well
- DMSO lot
- DMSO water uptake
- stock freeze–thaw
- serum lot
- media lot
- passage
- thaw date
- seeding density
- confluence at dosing
- dosing to fixation
- stain and dye lots
- operator
- incubator shelf
- edge evaporation
- mycoplasma status
- CO2 and humidity
- compound dose

not required to publish today
in our own data: tracked, every one

CAN A MODEL TELL WHICH LAB?
HIGHER = CLEARER LAB FINGERPRINT




29,943 DMSO wells, 16 per plate
tested on plates it never saw

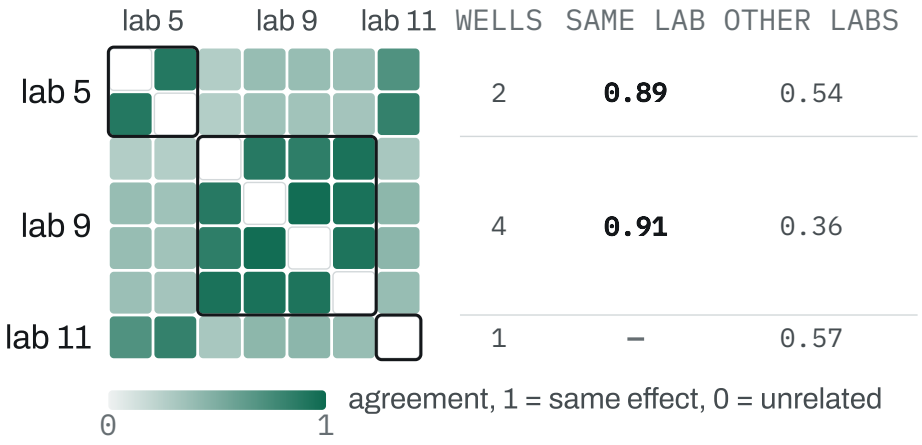
Batch correction is built to remove this signal [18], variables: JUMP metadata [7]

# The same drug looks less alike in another lab.

Darker = more alike.

WHAT A DRUG SCREEN REPORTS drug wells – DMSO wells, same plate

RSL3 AT 10 µM, EVERY WELL AGAINST EVERY OTHER




| | WELLS | SAME LAB | OTHER LABS |
|---|---|---|---|
| lab 5 | 2 | **0.89** | 0.54 |
| lab 9 | 4 | **0.91** | 0.36 |
| lab 11 | 1 | – | 0.57 |

SAME 7 PLATES, EACH WELL AGAINST ITS OWN PLATE'S DMSO

| | SAME LAB | OTHER LABS |
|---|---|---|
| RSL3, 10 µM | **0.90** | 0.49 |
| 8 known drugs, 5 µM, median | **0.74** | 0.26 |
| DMSO vs DMSO | **0.00** | 0.00 |

DOES IT CHANGE THE NUMBER?

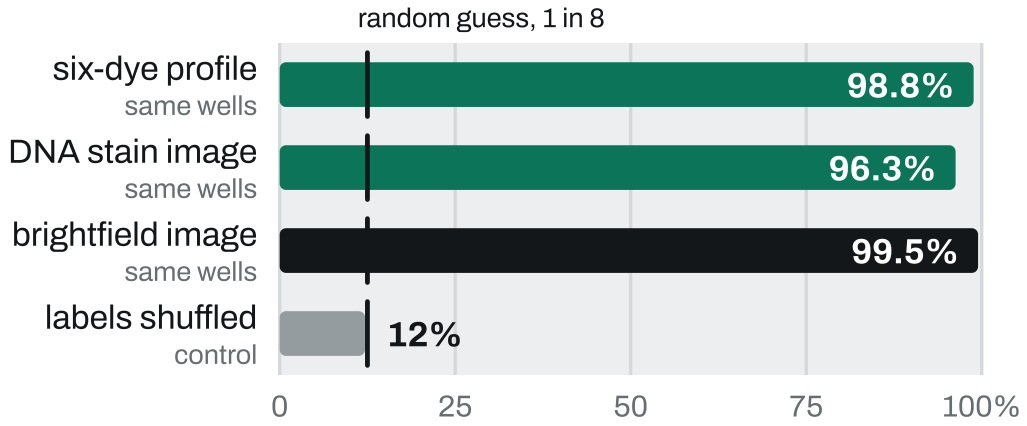
Which lab ran it explains 20% to 71% of how strong a drug's effect looks.

Seven wells: an example, not proof, lab 7, dosed at a sixteenth, left out, agreement = cosine of two wells' effects, each well minus its own plate's DMSO wells, 8 drugs at full scale: 1,829 plates, 11 labs, JUMP [7]

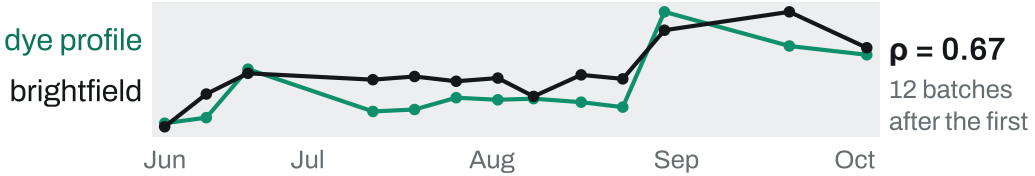
# One brightfield image already names the lab.

Evidence: six dyes.
Product: no dye.

CAN ONE WELL TELL WHICH OF 8 LABS IT CAME FROM?




DOES BRIGHTFIELD SEE LAB 2 DRIFT?, EACH LINE SCALED TO ITS PEAK

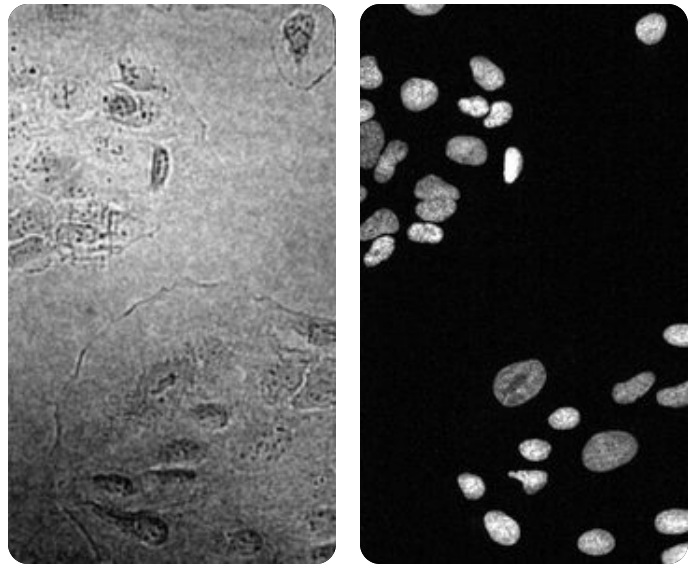



Not yet the whole map: beyond each lab's usual neighbour, brightfield and dye profile agree batch by batch no more than luck (45 of 102 against 41).

WHY THE BET IS FAIR

## No dye: fewer things can go wrong

~~stain lot~~ ~~dye lot~~ ~~staining step~~ A1



BRIGHTFIELD DNA STAIN

same spot in one well, lab 2

% named right, each lab counted equally, unseen batches, one DMSO well per plate, 1,338 plates, the same wells and model for every row, median of 20 splits of the batches, profile without its 8 frame-size columns, shuffled: 5 seeds per input, image model never trained on cells, JUMP [7]

# On the map, labs agree a little more on the same drug.

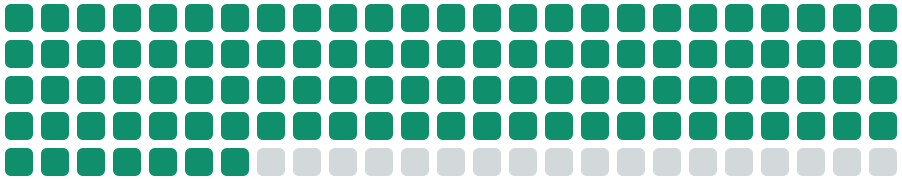
Seven known drugs, 125 batches.

## Do labs agree on what a drug does?

**0.75** → **0.76** → **0.78**

raw | usual fix | on the map

1 = every lab sees the same effect



**107 of 125 batches** higher on the map than with the usual fix

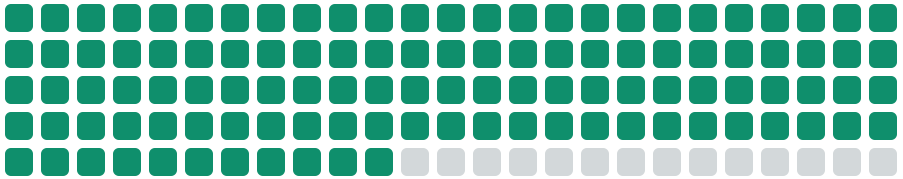
than raw: 121 of 125

## Can you still tell the drugs apart?

**0.52** → **0.56** → **0.59**

raw | usual fix | on the map

gap to the nearest wrong drug, higher is easier



**111 of 125 batches** higher on the map than with the usual fix

than raw: 125 of 125

### NOTHING LOST

Right drug recognised: 0.92 → 0.94

Matches its own past: 0.88 → 0.88

### HOLDS

Labs agree more: 5 of 5 ways of checking

Drugs stay apart: 4 of 5

### NOT FIXED YET, DRUG STRENGTH

Varies batch to batch, lower is better

raw 6% → usual fix 20% → map 12%: steadier than the usual fix, not yet as steady as raw

usual fix: each batch set against its own “normal”, its DMSO wells, map: against every other lab’s, 1,829 plates, JUMP [7]

proposed, prices are the assumptions on A10

# Labs earn access by uploading; AI teams pay the most, by API.

| WHO | WHAT THEY GET | WHAT THEY PAY | WHAT IT BUILDS |
|---|---|---|---|
| **labs and core facilities** | membership: the repository and the map; a check on every batch, the leaderboard | **nothing: uploading controls earns it** | the map |
| **teams that run cell screens all year** | membership: a check on every batch they run | **~$5–15k/line/quarter** | first revenue |
| **AI labs and model builders** | membership by API: aligned raw images from every lab | **the most: priced by use, not yet set** | main revenue |
| **companies that buy cell data** | membership: each dataset they buy, placed on the map | **~$10–50k/dataset** | data revenue |
| **benchmark partners** | membership: a test set with faults planted on purpose | **~$50–200k/year** | the standard |

only control wells leave a lab, DMSO and known drugs, never a test compound, privacy A6, the lab’s share A11

# Only control wells leave the lab.

Nothing for legal to approve.

## WHAT LEAVES THE BUILDING

**control wells only: DMSO and known drugs**



green: 64 of 384 wells leave, JUMP [7]

**images, or numbers from them**

proposed: images deleted once measured

**a plate map and the batch date**

which wells, and when they were imaged

## WHAT DATUM LINE DOES WITH IT

**builds the map**

scale set by the other labs, never its own

**the map holds DMSO wells only**

**82,319**

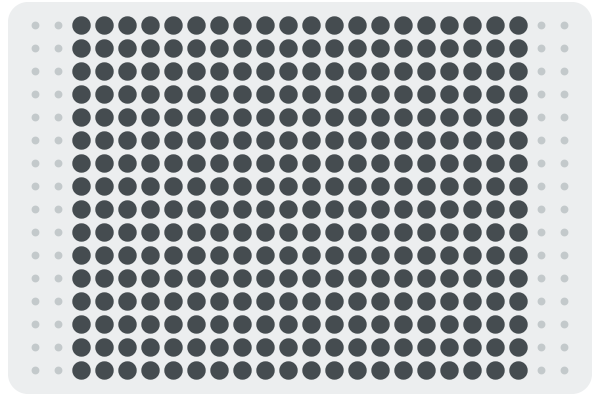
for lab 2: DMSO wells of 10 other labs
0 treated

**answers**

“normal” or not, how far it moved, nearest lab

## WHAT IT NEVER SEES

**your treated wells**



dark: 320 of 384 wells stay with you

**your results**

corrected in your building, by you

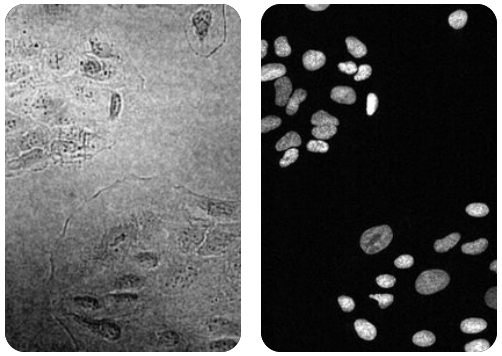
**your name, unless you choose**

proposed: a number until you opt in.
Every lab here is a number.

In every analysis here, the map and the correction see DMSO wells only, github.com/kat-titter/datum-line/analysis, JUMP [7]

# An image goes in; a position and a correction come out.

## 1, AN IMAGE



brightfield DNA stain

one DMSO well, JUMP [7]

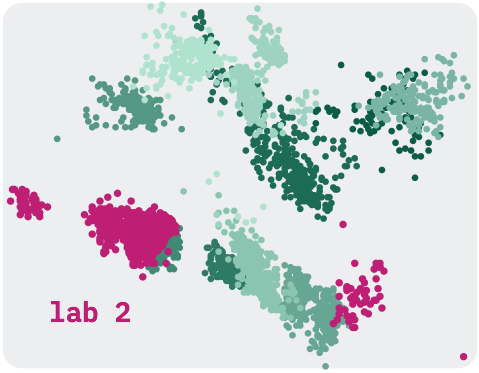
## 2, NUMBERS

**3,773**

numbers per well

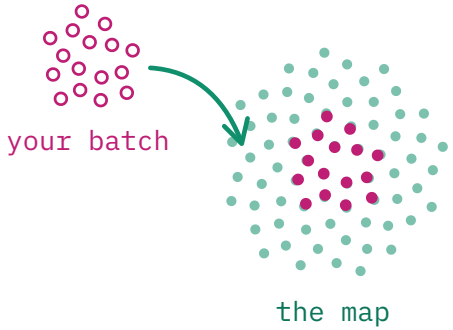
standard software today: size, shape, brightness, texture

## 3, THE MAP




each lab placed on the other labs' scale, never its own

## 4, THE CORRECTION




one per batch, fitted on your DMSO wells, sketch, A4

## AI IMAGE MODEL NEXT, A3

| | |
|---|---|
| brightfield, 8 labs | 99.5% |
| DNA stain, 8 labs | 96.3% |

right lab, from one picture

Correction, per batch on 200 components: $x' = C_{field}^{\frac{1}{2}} C_{batch}^{-\frac{1}{2}} (x - m_{batch})$, every script: github.com/kat-titter/datum-line/analysis

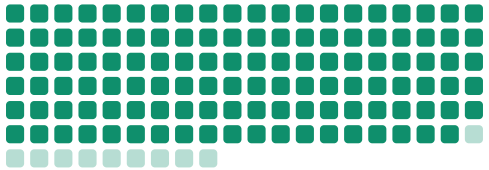
# Every lab leaves a fingerprint, so a wrong label stands out.

Two work today, three are proposed.

PROVENANCE today

## 99.9%

of 93,228 DMSO wells get the right lab, on unseen plates: cells and microscope together.



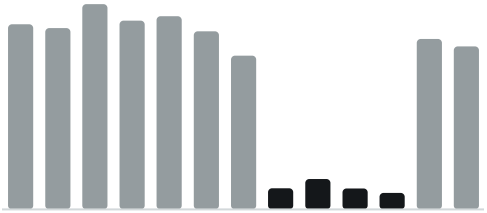
1 square = 1 batch
solid: every well right, 119

INTEGRITY today

## 4 batches

Label: known drug in these wells.

Cells: almost no effect, on 65 plates.

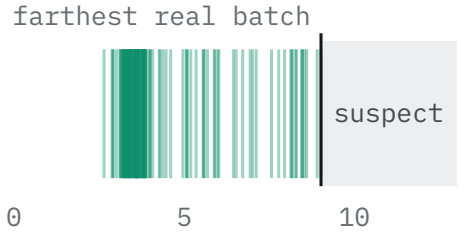


lab 3, drug effect per batch
dark: failed, archive A8

UNKNOWN ORIGIN proposed

## 9×

Farthest of 129 real batches from its nearest other lab.

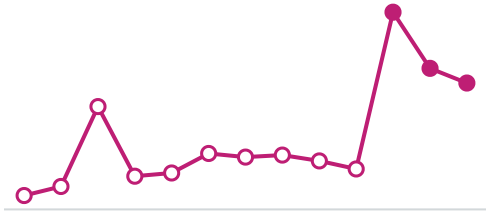



distance to nearest other lab
1 line = 1 real batch

EARLY WARNING proposed

## 30 Aug

2021: lab 2’s DMSO wells moved; its cell count did not.



lab 2, since batch 1
filled: flagged, archive A13

DECLARED VS OBSERVED

proposed, opt-in

A company that wants the good-actor mark sends treated wells and says what is in them. The map flags cells that look like a known toxic class [7]. Only what a lab sends is seen.

Integrity check uses each plate’s known-drug wells (archive A8), × = times the usual spread of wells within one plate, JUMP [7]

proposed, every line an assumption

# About $1.8M for eighteen months, mostly people.

$884k plan + $884k contingency = $1.77M asked



people 71% | benchmark 17% | other 12%

| | | |
|---|---|---|
| **leader, to run it** | 18 months | **$180k** |
| **scientific lead (me)** | 18 months | **$180k** |
| **ML engineer** | from month 7 | **$160k** |
| **payroll overhead** | 20% of salaries | **$104k** |
| **benchmark playground** | ~60 plates, planted faults, archive A16 | **$120k** |
| **partner-lab bench time** | 3 labs, controls they already image, reagent top-ups, shipping | **$30k** |
| **compute and storage** | images, models, hosting, API | **$45k** |
| **legal, IP, insurance, accounting** | incorporation to first contract | **$45k** |
| **travel and partner visits** | 2 benchmark partners, 3 labs | **$20k** |
| **planned** | sum | **$884k** |
| **+ contingency** | most things take 2–3× the time and money planned, we plan for 2× | $884k |
| **asked, 18 months** | | **$1.77M** |

NO IN-HOUSE LAB

Partner labs or a contract lab.
Own lab: costs more, proves less.

WHAT THE MONEY MUST SHOW

**1** three partner labs, the first outside JUMP, paid in this round **2** first paying member
**3** benchmark playground **4** model for any image → seed

salaries below market, leader and scientific lead paid, nothing here committed

# About $84M a year, if every likely buyer joined.

| assumption | low | base | high |
|---|---|---|---|
| organisations testing drugs on a few cell lines | 200 | 400 | 800 |
| cell lines each one uses per quarter | 2 | 4 | 8 |
| price per cell line per quarter, $k | 5 | 10 | 15 |
| organisations testing drugs on hundreds to thousands of cell lines | 10 | 25 | 50 |
| site licence per year, $k | 50 | 150 | 300 |
| **per-line checks and site licences, per year** | **$8.5M** | **$67.8M** | **$399M** |
| companies buying cell data they did not make | 50 | 150 | 300 |
| purchased datasets each places on the map per year | 2 | 4 | 8 |
| price per dataset membership, $k | 10 | 25 | 50 |
| **dataset memberships on purchased data, per year** | **$1M** | **$15M** | **$120M** |
| benchmark partners | 4 | 10 | 20 |
| membership per partner per year, $k | 50 | 100 | 200 |
| **benchmark membership, per year** | **$200k** | **$1M** | **$4M** |
| **total per year, if every buyer joined** | **$9.7M** | **$83.8M** | **$523M** |

BASE CASE, PER YEAR




↑ five-year target, a tenth, $8.4M

benchmark $1M ↑

WHY BUYERS PAY

A line checked late, in a screen I ran:
3 runs a week × 17 weeks × $50k ≈ $2.6M to redo.
A $25k dataset membership: about 1% of that.

prices proposed on A5, academic labs join free, they are the map, not the revenue

# The lab keeps its data and a share; anyone can check it.

No crypto token.

### CONTROL WELLS ONLY — today

no test compound, no target, no result
results never leave the lab

### LAB KEEPS ITS DATA — proposed

sharing is opt-in
withdraw any time

### LOW RISK BY DESIGN — today

cell-line images only
no human genetic data, no pathogen data

### SIGNED AND TIMESTAMPED — proposed

public record, anyone can check
dates cannot be changed, blockchain optional

### KNOWN MEMBERS — proposed

verified at sign-up, odd access flagged
access log open to members

### OPEN CODE AND DATA — today

code: MIT, on GitHub
data: JUMP, CC0; every number from a script

**lab shares** → **the map checks** → **members subscribe** → **lab keeps a share**

control images, opt-in
credited by name on the map

how “normal” the cells look
which lab, what changed

to aligned data
from every lab

of every membership
as long as its data is used